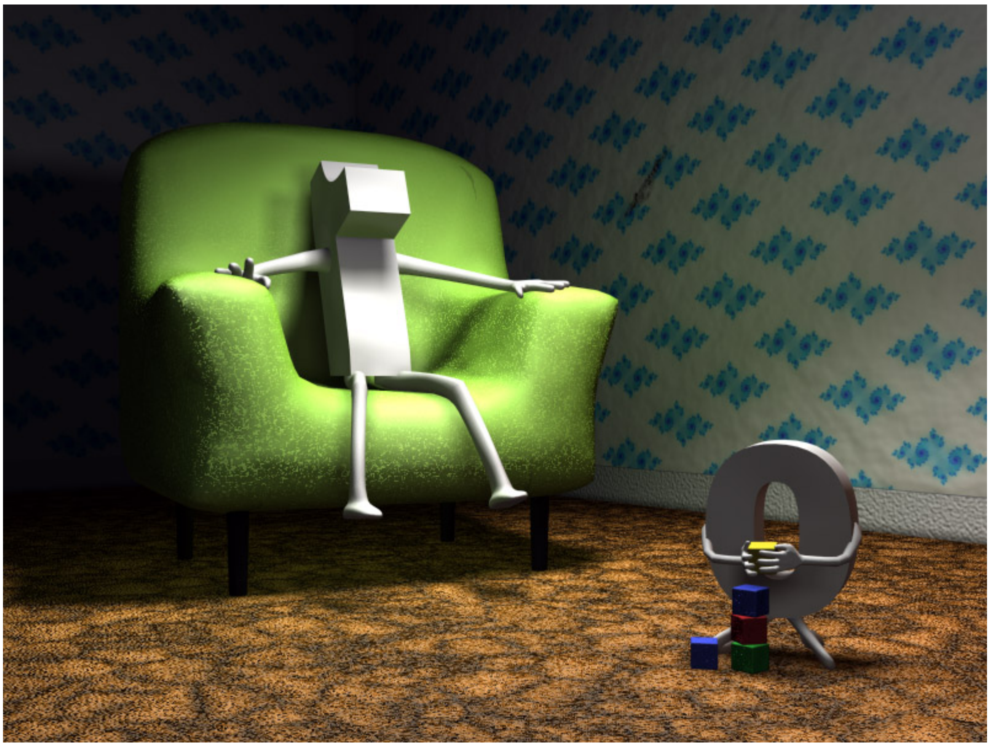
Quiz Week 2

Integers



Congruences

1. **In each case determine whether the statement is true or false.**

   (1) There is an integer x such that $13x \equiv 9 \pmod{35}$. If true find x.

   (2) $-29 \equiv 1 \pmod 7$

   (3) $132 \equiv 0 \pmod{11}$

   (4) $3^4 \equiv 1 \pmod 5$

   (5) The remainder when 10^{515} is divided by 7 is 6.

Solution: (1) is **true** because the integers 13 and 35 are coprime, which tells us that $\overline{13}$ has a (multiplicative) inverse in $\mathbb{Z}_{35}$, and we can therefore solve the (equivalent) equation $\overline{13}x = \bar{9}$ in $\mathbb{Z}_{35}$. So, we first find the inverse of $\overline{13}$ in $\mathbb{Z}_{35}$; to do so we apply the Euclidean algorithm. From

$$\begin{aligned} 35 &= 2 \cdot 13 + 9, \\ 13 &= 1 \cdot 9 + 4, \\ 9 &= 2 \cdot 4 + 1, \end{aligned}$$

we get that

$$\begin{aligned} 1 &= 9 - 2 \cdot 4 = 9 - 2 \cdot (13 - 1 \cdot 9) = (-2)13 + 3 \cdot 9 = \\ &= (-2)13 + 3(35 - 2 \cdot 13) = 3 \cdot 35 + (-8)13, \end{aligned}$$

which yields that $\overline{13}^{-1} = \overline{-8} = \overline{27}$ in $\mathbb{Z}_{35}$. To solve the equation $\overline{13}x = \bar{9}$ in $\mathbb{Z}_{35}$, multiply both sides of the equation by $\overline{27}$ and reduce (if necessary) modulo 35:

$$x = \overline{27} \cdot \bar{9} = \overline{-8} \cdot \bar{9} = \overline{-72} = \overline{-2} = \overline{33}.$$

(2) is **false** since $-29 - 1 = -30$ is not a multiple of 7.

(3) is **true** because $132 - 0 = 12 \cdot 11$, i.e., a multiple of 11.

(4) is **true** since $3^4 - 1 = 81 - 1 = 80 = 5 \cdot 16$, i.e., a multiple of 5.

(5) is **false**. In fact, let us compute the remainder when 10^{515} is divided by 7, or equivalently $\overline{10^{515}}$ in $\mathbb{Z}_7$. Keeping in mind that $\overline{10^{515}} = \overline{10}^{515}$, the idea is to find d such that $\overline{10}^{d} = \bar{1}$, and divide 515 by d. So let us do that! In $\mathbb{Z}_7$ we have that

$$\begin{aligned} \overline{10}^1 &= \bar{3}, \\ \overline{10}^2 &= \bar{3} \cdot \bar{3} = \overline{3 \cdot 3} = \bar{9} = \bar{2}, \\ \overline{10}^3 &= \overline{10}^2 \cdot \overline{10} = \bar{2} \cdot \bar{3} = \overline{2 \cdot 3} = \bar{6}, \\ \overline{10}^4 &= \overline{10}^3 \cdot \overline{10} = \bar{6} \cdot \bar{3} = \overline{6 \cdot 3} = \overline{18} = \bar{4}, \\ \overline{10}^5 &= \overline{10}^4 \cdot \overline{10} = \bar{4} \cdot \bar{3} = \overline{4 \cdot 3} = \overline{12} = \bar{5}, \\ \overline{10}^6 &= \overline{10}^5 \cdot \overline{10} = \bar{5} \cdot \bar{3} = \overline{5 \cdot 3} = \overline{15} = \bar{1}. \end{aligned}$$

So, $d = 6$ and $515 = 6 \cdot 85 + 5$, which implies that

$$\overline{10}^{515} = (\overline{10}^{6})^{85} \cdot \overline{10}^{5} = \overline{10}^{5} = \bar{5}.$$

Thus the remainder when 10^{515} is divided by 7 is 5.